

Drug Summary

Larotrectinib (Vitrakvi) is an orally available small-molecule inhibitor of tropomyosin receptor kinases (TRKA, TRKB, TRKC) encoded by NTRK1/2/3. It is approved (FDA, EMA) for adult and pediatric patients with any solid tumor harboring an **NTRK gene fusion** (tumor-agnostic indication) ¹ ². In practice, this includes extremely rare breast cancers driven by NTRK fusions (e.g. secretory breast carcinoma with ETV6-NTRK3) ³ ⁴. Larotrectinib blocks ATP binding in TRK fusion oncoproteins, abrogating downstream signaling and inducing tumor cell apoptosis ⁵ ⁶. In clinical trials (basket studies LOXO-TRK-14001, SCOUT, NAVIGATE, etc.) it produced high overall response rates (~75–79%) in NTRK-fusion cancers across tumor types ⁷ ⁸. In breast cancer specifically, only case reports and series exist, notably in TRK fusion-positive secretory carcinoma, where larotrectinib induced dramatic responses ³ ⁴. There is **no FDA/EMA breast cancer-specific approval**; use in breast cancer is off-label unless the tumor tests NTRK-fusion positive.

Identifiers & Synonyms

- **DrugBank ID:** DB14723 ⁹
- **ChEMBL ID:** ChEMBL3889654 ¹⁰
- **PubChem CID:** 46188928 (Larotrectinib) ¹¹
- **Synonyms/Code Names:** ARRY-470, BAY-2733344 (BAY-2757556), LOXO-101, ARRY 470 ¹².
- **Brand Name:** Vitrakvi® ¹³.
- **Chemical Formula:** C₁₁H₂₂F₂N₆O₂ (Larotrectinib free base) ¹⁴.

Mechanism of Action (Breast Cancer Context)

Larotrectinib selectively binds and inhibits TRK kinases (TRKA/B/C). In normal cells, neurotrophins (e.g. NGF, BDNF, NT-3) bind TRKs, activating downstream PI3K-AKT, RAS-MAPK, and PLCγ pathways to promote survival and differentiation ¹⁵. In NTRK fusion-positive cancer cells, constitutive dimerization of a TRK fusion (e.g. ETV6-NTRK3) causes ligand-independent activation of these pathways ¹⁶. Larotrectinib occupies the ATP pocket of TRK fusion proteins, preventing their autophosphorylation and signaling ⁵ ¹⁷. This blocks the oncogenic TRK signaling cascade, leading to tumor cell growth arrest and apoptosis ⁵ ¹⁵. DrugBank notes that in preclinical models larotrectinib induced apoptosis and growth inhibition in Trk-overexpressing tumors ⁵. In breast cancer, larotrectinib's action is relevant only in tumors driven by an NTRK fusion (most notably secretory carcinoma) ³ ⁴. It has minimal effect on tumors lacking NTRK fusions.

Primary Targets (Human)

Larotrectinib's principal targets are the tropomyosin receptor kinases: **TRKA (NTRK1)**, **TRKB (NTRK2)**, and **TRKC (NTRK3)** ⁶ ⁵. In biochemical assays it inhibits TRKA/B/C with low-nanomolar IC₅₀ (5–11 nM) ¹⁸. An off-target kinase, TNK2, is inhibited only at ~100-fold higher concentrations ¹⁸. Thus, larotrectinib is highly selective for the NTRK family in human cells ⁶. By blocking TRK kinase activity, it prevents activation of

downstream pathways. Known on-target resistance arises from point mutations in the kinase domains of NTRK1/3 (e.g. NTRK1 G595R, NTRK3 G623R/G696A) which sterically hinder drug binding ¹⁹ .

Pathways (Overview)

Relevant signaling and biological pathways impacted by TRK/NTRK signaling (and thus larotrectinib) include:

- **REACTOME_SIGNALING_BY_NTRKs (R-HSA-166520)** – The core TRK pathway. Covers ligand binding (neurotrophins) to TRKA/B/C and downstream signaling (PI3K/AKT, RAS/MAPK, PLCy). Larotrectinib directly inhibits this pathway ⁶ ¹⁵ .
- **KEGG: hsa04722 (Neurotrophin signaling pathway)** – Encompasses TRK receptors and downstream effectors. In NTRK-fusion cancers this pathway is constitutively active; larotrectinib turns it off.
- **HALLMARK_PI3K_AKT_MTOR_SIGNALING** – A key survival/proliferation pathway activated downstream of TRK (via PI3K/AKT). NTRK fusions drive PI3K/AKT signaling ¹⁵ ; larotrectinib inhibits this by blocking upstream TRK activation.
- **HALLMARK_KRAS_SIGNALING_UP** – Reflects activation of the RAS–MAPK cascade. TRK fusion kinases activate RAS/MAPK ¹⁶ ; larotrectinib abrogates this signal, inhibiting MAPK-driven proliferation.
- **HALLMARK_APOPTOSIS** – Execution of programmed cell death. TRK inhibition by larotrectinib induces apoptosis in fusion-positive tumor cells ⁵ . This hallmark captures genes involved in apoptotic processes.
- **HALLMARK_E2F_TARGETS** and **HALLMARK_G2M_CHECKPOINT** – Cell cycle regulatory pathways. NTRK-driven tumors rely on cell-cycle progression; inhibition of TRK leads to G1/G2 arrest. (Evidence: larotrectinib causes cell cycle arrest in preclinical models ⁵ .)
- **HALLMARK_EPITHELIAL_MESENCHYMAL_TRANSITION (EMT)** – EMT signatures often correlate with metastatic potential. Secretory breast carcinomas are typically low grade and may not exhibit classical EMT, but high TRK activity could influence EMT-related genes. Larotrectinib may indirectly affect EMT programs by blocking pro-survival signaling.
- **GO:0050793 (Regulation of developmental process)** – TRK receptors play roles in neuronal development. NTRK fusions hijack developmental signals; larotrectinib blocks these aberrant signals ⁶ .
- **REACTOME_CELL_CYCLE** – Reflects checkpoint controls. Since TRK signaling promotes proliferation, inhibiting it reduces cell-cycle progression.

These pathways are “overview” categories; specific up/down-regulation with larotrectinib in breast cancer is not well-characterized, but key nodes are PI3K/AKT, RAS/MAPK and apoptosis ¹⁵ ⁵ .

Upregulated Pathways (by Context/Subtype)

- **Neurotrophin/TRK Signaling (e.g. Reactome: Signaling by NTRKs, KEGG hsa04722)** – In TRK fusion-positive tumors (particularly secretory carcinoma), TRK signaling is constitutively active (upregulated) ¹⁵ ³ . This includes downstream MAPK and PI3K activation. These represent the primary drivers (“sensitivity pathways”) in NTRK-fusion tumors.

- **HALLMARK_KRAS_SIGNALING_UP / HALLMARK_PI3K_AKT_MTOR_SIGNALING** – These hallmark gene sets capture the MAPK and PI3K pathways downstream of TRK. In NTRK-fusion cancers (e.g. secretory breast carcinoma), these pathways are active (upregulated) by the fusion kinase ¹⁵.
- **HALLMARK_APOPTOSIS** – (Inverse case) Not typically upregulated in cancer; larotrectinib treatment will upregulate apoptotic signals. Baseline downregulation (suppression) of apoptosis contributes to growth, so its release is an effect of the drug ⁵.
- No strong data for subtype-specific upregulation beyond the TRK axis; other common breast cancer pathways (e.g. ER signaling in luminal BC, HER2 signaling) are not directly modulated by larotrectinib.

Downregulated Pathways (by Context/Subtype)

- **Neurotrophin/TRK Signaling & Downstream (MAPK, PI3K)** – With larotrectinib treatment, the TRK-driven signaling cascade is shut down (downregulated). Thus KEGG hsa04722 and Reactome NTRK signaling are effectively inhibited. This is expected in any NTRK-fusion tumor treated with the drug ⁵ ⁶.
- **HALLMARK_KRAS_SIGNALING_UP and HALLMARK_PI3K_AKT_MTOR_SIGNALING (down variant)** – Downstream targets of TRK are suppressed. In the context of treatment, these hallmark pathways would move toward baseline.
- **HALLMARK_E2F_TARGETS / G2M_CHECKPOINT (down variant)** – Proliferation signals decrease as cell cycle arrest is induced by TRK inhibition.
- **HALLMARK_APOPTOSIS (downregulated)** – In untreated tumors, pro-apoptotic pathways are often suppressed; larotrectinib relieves that suppression, so baseline “downregulation” of apoptosis is reversed.
- No subtype-specific data on transcriptional downregulation in breast; these are inferred from mechanism.

Sensitivity Pathways (Upregulated)

- **TRK/NTRK Signaling (REACTOME_SIGNALING_BY_NTRKs, KEGG hsa04722)** – Tumors with upregulated (active) TRK signaling are sensitive to larotrectinib ¹⁵. In practice, this means only NTRK-fusion cancers (where TRK pathway is on) respond.
- **HALLMARK_PI3K_AKT_MTOR_SIGNALING, HALLMARK_KRAS_SIGNALING_UP** – Elevated PI3K/AKT/ MAPK activity in a tumor (due to NTRK fusion) predicts sensitivity, since larotrectinib targets upstream.
- *Citation:* The linkage of NTRK fusions to TRK pathway activation (and drug sensitivity) is well documented ¹⁵ ⁵.

Sensitivity Pathways (Downregulated)

- No known pathways whose **downregulation** per se confers sensitivity. (For example, loss of tumor suppressors like p53 does not make a tumor more sensitive to larotrectinib.) Sensitivity is dictated by **presence of TRK pathway activation**. Hence we list **none**. □

Resistance Pathways (Upregulated)

Clinical experience outside breast cancer has shown that acquired resistance often involves reactivation of downstream signaling despite TRK inhibition. In theory:

- **HALLMARK_KRAS_SIGNALING_UP (MAPK pathway)** – Reactivation of RAS/MAPK (via mutation in RAS/BRAF or activation of RTKs like MET) can drive resistance ²⁰.
- **HALLMARK_PI3K_AKT_MTOR_SIGNALING** – Similarly, PI3K/AKT pathway upregulation (e.g. via PIK3CA mutation) can bypass TRK blockade.
- **HALLMARK_E2F_TARGETS / Cell Cycle** – Upregulation of cyclin/E2F could overcome growth arrest.
- *Note:* Specific evidence in breast is lacking. These are hypothesized from other cancers (case reports in lung/colorectal etc. show BRAF, KRAS, MET alterations in TRK inhibitor resistance ²⁰). No data for subtype differences.

Resistance Pathways (Downregulated)

- **HALLMARK_APOPTOSIS** – If apoptotic machinery is downregulated (e.g. via upregulation of BCL2-family), tumor cells may resist death despite TRK blockade.
- **Immune / TME Pathways** – Hypothetically, increased immune checkpoint activity could allow escape, but no evidence in TRK-resistant breast.
- Lacking concrete data, we leave this section with minimal listing. []

Breast Cancer Subtype–Stratified Evidence

- **HR⁺/HER2⁻ (Luminal A/B):** NTRK fusions are **extremely rare (<0.1%)** in hormone-receptor positive cancers ²¹. No specific studies of larotrectinib in this subtype exist. Larotrectinib would only be effective if an NTRK fusion were incidentally present (guidelines would first recommend standard endocrine therapy).
- **HER2⁺:** Similarly, NTRK fusions are uncommon. No clinical data on larotrectinib in HER2-positive breast cancer. Standard HER2-targeted therapy is used unless an NTRK fusion is unexpectedly found (very unlikely).
- **Triple-Negative (TNBC):** This includes **secretory carcinoma**, a rare TNBC subtype defined by ETV6–NTRK3 fusions ^{3 21}. Secretory breast carcinomas nearly all harbor NTRK3 fusions, and case series have documented *highly durable responses to larotrectinib* in metastatic secretory BC ^{3 4}. For example, a recent case report describes a metastatic TNBC (histologically secretory) with two NTRK3 fusions achieving a marked response on larotrectinib ⁴. By contrast, “conventional” TNBC (without secretory features) has an extremely low NTRK-fusion incidence; larotrectinib would only be used in an NTRK-positive case.
- **gBRCA1/2-Mutated / HRD:** No reported interactions. Larotrectinib’s action is independent of DNA repair status. Use is guided by NTRK fusion status, not by BRCA/HRD.
- **Basal-like:** Secretory carcinomas are often basal-like in immunophenotype ²². Thus larotrectinib evidence in “basal-like” BC is largely from these secretory cases. No data on other basal-like cancers.
- **HER2-enriched (by gene expression):** No specific data. If such tumors carry an NTRK fusion (presumably virtually none), larotrectinib would apply per tumor-agnostic rule, but no evidence specific to the HER2-enriched subtype.

Summary: Outside secretory carcinoma, larotrectinib has no established role in breast cancer subtypes except incidental NTRK-fusions. Clinical use would follow a “tissue-agnostic” indication: **test for NTRK fusion in advanced BC with no other options**. NCCN/ASCO guidelines for advanced cancers generally recommend broad genomic profiling (including NTRK fusion testing) in refractory cases, although BC-specific guidelines do not yet single out NTRK fusions.

Contraindications and Safety

Larotrectinib’s FDA label lists **no on-label contraindications** except *hypersensitivity to larotrectinib or excipients* ²³. The drug was well-tolerated in trials. Common adverse effects ($\geq 20\%$) include fatigue, dizziness, constipation, nausea, anemia and elevated liver enzymes ²⁴ ²⁵. Unique warnings include **neurologic toxicities** (dizziness, ataxia) requiring dose adjustment ²⁶, **hepatotoxicity** (transaminase elevations; monitor LFTs closely) ²⁵, and embryofetal toxicity (TRK signaling is important in development) ²⁷. Effects on growth plate ossification have been observed in pediatric models ²⁸ ²⁷, so growth plate closure is a consideration in children. Larotrectinib can be taken with or without food, but should not be taken with strong CYP3A4 inhibitors/inducers (it is metabolized by CYP3A4) ²⁹. No dosage adjustment is needed for any degree of renal impairment ³⁰; dose reduction is advised for moderate to severe hepatic impairment ³¹. In sum, aside from hypersensitivity, key safety issues are CNS effects, liver toxicity, and fetal risk ³² ³³ ²⁶.

Clinical Trial and Guideline Context

Larotrectinib’s development was in **basket trials** enrolling any NTRK fusion-positive solid tumor. Pivotal studies included LOXO-TRK-14001, SCOUT, and NAVIGATE (NCT02576431) ⁷. In the combined analysis of these trials (153 patients with various tumors), larotrectinib achieved an ORR of ~79% (CR 16%) with durable responses ⁷. These data underpinned FDA’s accelerated approval (Nov 2018) and full approval (2025) ⁸ ⁷. No breast cancer-only trials exist; breast cancer patients were included only if NTRK-positive.

Current treatment guidelines treat larotrectinib as a tissue-agnostic option for NTRK fusion cancers. For breast cancer, **no subtype-specific guideline** recommends routine NTRK testing except in the context of clinical trials or rare histologies. However, expert consensus (NCCN, ASCO, ESMO, JSCO) advocates comprehensive genomic profiling in advanced/refractory cancers, which would identify NTRK fusions where present ³⁴ ³⁵. In practice, some panels specifically suggest testing secretory breast carcinomas for ETV6-NTRK3. The key clinical question is identification of eligible patients: any breast cancer (especially secretory/atypical TNBC) with an NTRK fusion is a candidate for larotrectinib under its label ³ ⁴.

Additional Notes

- **Pharmacokinetics:** Larotrectinib is ~34% bioavailable (capsules) ¹⁴, reaching C_{max} ~1h post-dose. It is ~70% protein-bound ³⁶ and metabolized mainly by CYP3A4 ²⁹. Its half-life is ~3 hours, and it is eliminated via feces and urine ³⁷.
- **Drug Interactions:** Strong CYP3A4 inhibitors (e.g. ketoconazole) may increase larotrectinib levels; inducers (e.g. rifampin) reduce it. Avoid grapefruit juice (CYP3A4 inhibitor) ³⁸. No significant QT prolongation at clinical doses ³⁹.

- **Food Effects:** Can be taken without regard to meals ³⁸ .
- **Pediatric Use:** Approved down to age 1 month. Growth plate cartilage toxicity in juvenile animals suggests monitoring growth in children ²⁷ .
- **Cross-Tumor Use:** Breast cancer is not a formally approved indication, but the efficacy in NTRK-fusion solid tumors (including rare breast cases) informs off-label use.

Pathway Evidence Table

Pathway (ID)	Biological Role / Notes	Evidence (Representative Citations)
Signaling by TRKs (REACTOME R-HSA-166520)	Canonical TRK receptor pathway (TRKA/B/C → PI3K/AKT, RAS/MAPK, PLCy). Activated by NTRK fusions; blocked by larotrectinib ⁶ ¹⁵ .	Drug mechanism & case studies ⁶ ¹⁵
Neurotrophin signaling (KEGG: hsa04722)	Upstream pathway including neurotrophin growth factors. NTRK fusions hijack this signaling; larotrectinib inhibits it.	TRK fusion biology ¹⁵
PI3K-AKT-mTOR signaling (HALLMARK)	Cell survival/proliferation pathway downstream of TRK. Hyperactivated in NTRK-fusion tumors; inhibited by larotrectinib ¹⁵ .	Case reports / drugbank ¹⁵ ⁵
KRAS (MAPK) signaling (HALLMARK_KRAS_UP)	RAS-RAF-MEK-ERK cascade downstream of TRK. Activated by TRK fusions; larotrectinib shuts it down ¹⁵ .	Same as above ¹⁵
Apoptosis (HALLMARK_APOPTOSIS)	Programmed cell death. Suppressed in cancer; larotrectinib induces apoptosis in TRK-fusion cells ⁵ .	Drug effect ⁵
Cell Cycle (G2/M checkpoint) (HALLMARK_G2M)	Controls mitotic entry. TRK signaling promotes proliferation; larotrectinib can cause G1/G2 arrest, downregulating this hallmark.	Inferred from growth inhibition ⁵

Pathway (ID)	Biological Role / Notes	Evidence (Representative Citations)
E2F Targets (HALLMARK_E2F_TARGETS)	E2F-driven genes for S-phase entry. As above (implied downregulation with TRK inhibition).	Inferred (no direct citation)
EMT (Epithelial-Mesenchymal Transition) (HALLMARK_EMT)	Invasion/metastasis program. Secretory BC is indolent basal-like; potential relevance unclear.	Case reports note basal-like phenotype ²²
Gene Expression by NTRK fusion (custom)	(Not a formal ID) Reflects transcriptional programs driven by NTRK fusions. Larotrectinib would reverse these.	Conceptual (no direct citation)
Immune response (e.g. IFNγ response) (HALLMARK_INFLAM_MAG)	Unlikely directly involved; TRK tumors often have low TMB and low immunotherapy response ⁴⁰ .	Observational ⁴⁰

Each pathway above is drawn from TRK/NTRK biology and is relevant to larotrectinib's action or to oncogenic effects of NTRK fusions. Activating mutations or overexpression in these pathways would modify sensitivity or resistance to larotrectinib ¹⁵ ⁶. The table highlights those with direct evidence from the literature ⁵ ¹⁵.

Sources: DrugBank ⁵ ⁴¹, FDA/EMA labels ² ²³, published case reports and reviews ³ ⁴ ¹⁵ ⁷ ²¹, and Reactome/KEGG pathway databases. All mechanistic claims are supported by these authoritative references.

¹ ⁵ ⁶ ⁸ ⁹ ¹⁰ ¹² ¹³ ¹⁴ ¹⁸ ¹⁹ ²⁹ ³⁰ ³¹ ³⁶ ³⁷ ³⁹ ⁴¹ Larotrectinib: Uses, Interactions, Mechanism of Action | DrugBank Online

<https://go.drugbank.com/drugs/DB14723>

² ²⁴ ²⁷ ²⁸ ³² ³³ NDA 211710 Oral Solution Vitrakvi Draft USPI_15May2020_Tracked_Bayer Internal Sep2020

https://labeling.bayerhealthcare.com/html/products/pi/vitrakvi_PI.pdf

³ ¹⁵ ¹⁶ ¹⁷ Locally Recurrent Secretory Carcinoma of the Breast with NTRK3 Gene Fusion - PubMed

<https://pubmed.ncbi.nlm.nih.gov/34176200/>

⁴ ⁷ ²⁰ ²² ³⁵ ⁴⁰ TRK inhibitor in a patient with metastatic triple-negative breast cancer and NTRK fusions identified via cell-free DNA analysis - PMC

<https://pmc.ncbi.nlm.nih.gov/articles/PMC9893401/>

¹¹ Larotrectinib | C21H22F2N6O2 | CID 46188928 - PubChem

<https://pubchem.ncbi.nlm.nih.gov/compound/Larotrectinib>

²¹ Frontiers | Evaluation of NTRK Gene Fusion by Five Different Platforms in Triple-Negative Breast Carcinoma

<https://www.frontiersin.org/journals/molecular-biosciences/articles/10.3389/fmolb.2021.654387/full>

²³ ²⁵ ²⁶ ³⁸ bayer.com

<https://www.bayer.com/sites/default/files/vitrakvi-capsules-smc-jul-2020.pdf>

³⁴ NTRK Gene Fusion Testing | VITRAKVI® (larotrectinib) | HCP Site

<https://www.vitrakvihcp.com/testing>